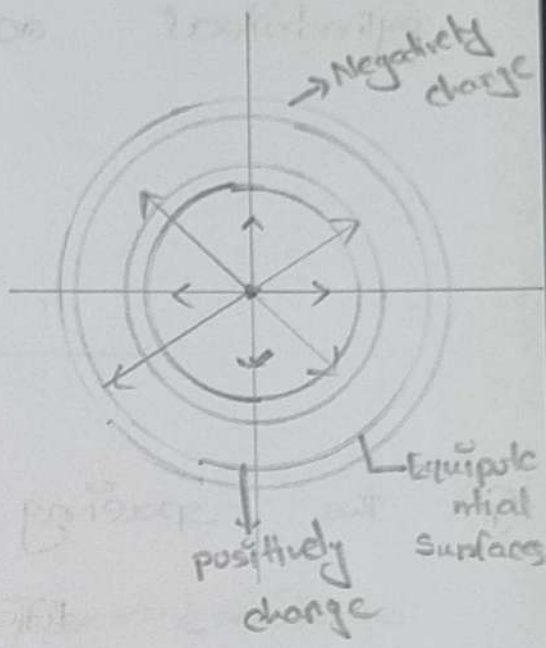
Ans. to the que. no.1

① Direction

The electric field inside the Geiger tube is directly radially outward from the central wire to the outer cylinder. This is because the central wire is positively charged, and electric field lines emanate from positive charges toward negative charges.



Higher electric potential

The central wire is at a higher electric potential since it carries a positive charge. In electrostatic systems, potential decreases with distance from a positive charge.

Shape

The equipotential surfaces are coaxial cylindrical surfaces centered on the central wire. These surfaces are concentric with the two cylindrical components of the Geiger tube.

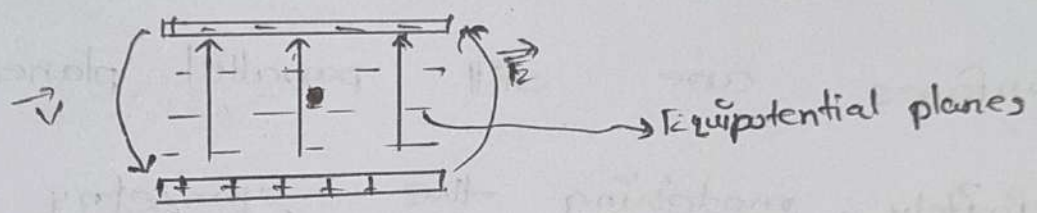
(ii) The spacing between two equipotential surfaces differing by 10V is not uniform. Near the central wire (small radius, r), the electric field is stronger ($E \propto 1/r$), so a smaller radial distance (Δr) is required to achieve a 10V difference. Conversely, near the outer cylinder (large r), the weaker field requires a larger Δr for the same 10V difference.

Equipotential surfaces are more widely spaced near

near the outer cylinder. This is confirmed by the relationship $\Delta V = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{r_2}{r_1}\right)$ where a fixed ΔV results in larger Δr at larger r.

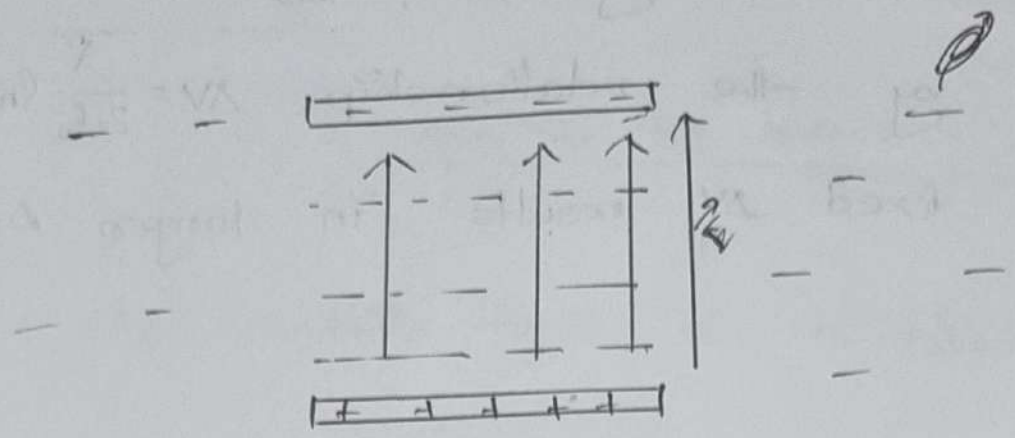
Ans. to the que. no. 2

(1)



The electric field between two parallel, oppositely charged plates is uniform. Equipotential surfaces are always perpendicular to the electric field. Since the field lines are straight and parallel, the equipotential surfaces are parallel planes oriented perpendicular to the field.

(11)



Outside the plates, the electric field is approximately zero (assuming ideal infinite planes). In this region, the potential remain constant. Thus, the equpotential surfaces are still parallel planes extending infinitely, matching the symmetry of the plates. However, ~~the~~ if the plates are finite, the equpotential surfaces would curve slightly at the edges, but this is negligible for large plates.

Ans. to the que. no. 3

(1) The electric field of a point charge is,

$$E = \frac{Q}{4\pi\epsilon_0 x^2}$$

Given,

$E = c/x^2$ where,

$$\frac{Q}{4\pi\epsilon_0} = c$$

$$Q = 4\pi\epsilon_0 c$$

Now,

$$Q = 4.\pi \times 8.854 \times 10^{-12} \times 10^{4}$$

$$= 1.11 \times 10^{-6}\ C$$

$$= 1.11\ \mu C$$

The electric field points radially away from the charge (since E is positive along +x). Thus, the charge is positive.

(iv) Given,

$x_1 = 1.m$

$x_2 = 2m$

potential difference,

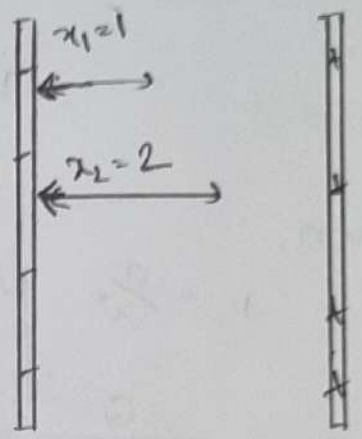
$$\Delta V = -\int_1^2 E dx$$

$$= -\int_1^2 \frac{c}{x^2} dx$$

$$= -c\left[-\frac{1}{x}\right]_1^2$$

$$= c\left(\frac{1}{2} - 1\right)$$

$$= -\frac{c}{2}$$



Now,

$$\frac{\Delta V}{2} = \frac{-10^4}{2}$$

$$= -5000V$$

This, implies $V(1) - V(2) = 5000V$. Thus, $x = 1.m$ has a higher potential than $x = 2$ m.

Ans. to the que. no. 4

(i) The midpoint is 0.25m from each ball. So, the total potential is the sum of potentials due to both charges,

$$V_{mid} = \frac{Q_1}{4\pi\epsilon_0 r} + \frac{Q_2}{4\pi\epsilon_0 r}$$

$$= \frac{9\times10^9}{0.25}\left(2\times10^{-9} - 1\times10^{-9}\right)$$

$$= 12\,V$$

Given,
$Q_1 = 2nC$
$Q_2 = 1nC$
$r = 0.25m$

Ans.

(ii) For ball 1,

At $r = 2cm$ from its center ($R_1 = 2.5cm$)

Potential due to ball 1,

$$V_{self} = \frac{kQ_1}{2R_1}\left(3 - \frac{r^2}{R_1^2}\right)$$

$$= \frac{9\times10^9\times2\times10^{-9}}{2\times0.025}\left(3 - \frac{(0.02)^2}{(0.025)^2}\right)$$

$$= \cancel{898.471}\ 899.6V$$

potential due to ball 2,

distance from ball 2 center $= 1.5 - 0.02 = 1.48\,m$

$$V_2 = \frac{kQ_2}{1.48} = -6.08V$$

$$\therefore V_{i_1} = 849.6 - 6.08$$

$$= 843.52V$$ Ans.

for ball 2,

$r = 2cm$

potential due to ball 2,

$$V_{self} = \frac{kQ_2}{R_2} = \frac{(9\times10^9)(-1.\times10^{-9})}{0.025}$$

$$= -360V$$

potential due to ball 1,

$$V_2 = \frac{kQ_1}{1.48}$$

$$= 12.16V$$

$$\therefore V_{i_2} = -360 + 1216$$

$$= -347.84V$$ Ans.

Ans. to the que. 5

(1)

We know,

$$R = \rho \frac{L}{A}$$

Given,

$L_F = 1.2m$

$L_n = 8m$

$A_n = A_F$

$\rho_n = 6.99 \times 10^{-8} \Omega m$

$\rho_F = 9.71 \times 10^{-8} \Omega m$

for iron,

$$R_F = \frac{\rho_F L_F}{A}$$

$$= \frac{9.71 \times 10^{-8} \times 1.2}{A}$$

$$= \frac{1.1652 \times 10^{-7}}{A} \Omega$$

for nikel,

$$R_n = \frac{\rho_n L_n}{A}$$

$$= \frac{6.99 \times 10^{-8} \times 8}{A}$$

$$= \frac{5.592 \times 10^{-7}}{A} \Omega$$

Ans.

(ii) We know,

$$V = IR$$

for Iron,

$$V_F = 1.5 \times \frac{1.1652 \times 10^{-2}}{A}$$

$$= \frac{1.7478 \times 10^{-2}}{A} \ V$$

Given,

$I = 1.5 A$

for nikel,

$$V_n = 1.5 \times \frac{5.592 \times 10^{-2}}{A}$$

$$= \frac{8.388 \times 10^{-2}}{A} \ V$$

Ans.

(iii) We know,

$$E = \frac{V}{L}$$

- for Iron,

$$E_F = \frac{V_F}{L_F}$$

$$= \frac{1.7478 \times 10^{-7}}{1.2\ A}$$

$$= \frac{1.4565 \times 10^{-7}}{A}\ Vm^{-1}$$

- for Nikel,

$$E_N = \frac{V_N}{L_N}$$

$$= \frac{8.388 \times 10^{-7}}{8A}$$

$$= \frac{1.0485 \times 10^{-7}}{A}\ Vm^{-1}$$

The electric field ~~is uniform~~ arises from the potential difference maintained across the wires by the current source. It drives the current through the wires.

The electric field is uniform within each wire because the wires are homogeneous and have constant cross-sectional areas. Ans